SRINIVAS UNIVERSITY

INSTITUTE OF ENGINEERING & TECHNOLOGY

MUKKA, MANGALURU-574146

MINI PROJECT REPORT

ON

Online Bookstore

Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN

COMPUTER SCIENCE & ENGINEERING

Submitted By,

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Department of COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that the Technical Seminar entitled ***"Online Bookstore"*** is a bonafide work carried out by ***Megha Mahabaleshwar Naik, Mithali H gatty*** bearing the **USN** ***01SU23CS104, 01SU23CS105*** in the partial fulfillment for the award of **Bachelor of Technology** in **Computer Science & Engineering** of the **Srinivas University- Institute of Engineering & Technology** during the year **2025-26.** It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The mini project report has been approved as it meets the academic criteria outlined for the said degree.

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DECLARATION

We ***Megha Mahabaleshwar Naik, Mithali H Gatty*** the student of 6th semester, **Bachelor of Technology** in **Computer Science & Engineering** of the **Srinivas University-Institute of *Online Bookstore*"** has been successfully completed by us in partial fulfillment of the requirements for the award of degree in **Bachelor of Technology** in **Computer Science & Engineering of Srinivas University- Institute of Engineering & Technology** and no part of it has been submitted for the award of degree or diploma in any university or institution previously.

Date: 28-03-2026
Place: Mukka

ABSTRACT

The **IBM Online Book Store Using Microservices** is a modern web-based application designed to provide an efficient and scalable platform for managing online book sales. The system is developed using microservices architecture, where different functionalities such as user management, book catalog, order processing, and notifications are implemented as independent services.

The primary objective of this project is to overcome the limitations of traditional monolithic systems by ensuring better scalability, flexibility, and maintainability. Each microservice is developed using Node.js and Express.js and communicates through RESTful APIs, enabling smooth and efficient interaction between services. MongoDB is used as the database to store and manage application data, while Docker is used for containerization and deployment.

The application allows users to browse books in the online book store, register and log in securely, place orders, and track their purchases. Administrators can manage book inventory and monitor system activities. The use of microservices ensures that each component can be developed, deployed, and scaled independently without affecting the overall system.

This project demonstrates the real-world implementation of an online book store using distributed system design and provides a strong foundation for building scalable e-commerce applications. It highlights the importance of modular architecture in modern software development and offers opportunities for future enhancements such as real-time notifications and cloud deployment.

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Chapter-1

INTRODUCTION

The rapid growth of e-commerce has transformed the way people purchase products, including books. An **online book store** provides users with the convenience of browsing, selecting, and purchasing books from anywhere at any time. However, traditional systems used for developing such platforms are often monolithic in nature, making them difficult to scale, maintain, and update.

The **IBM Online Book Store Using Microservices** is designed to address these challenges by adopting a modern microservices architecture. In this approach, the application is divided into multiple independent services such as user management, book catalog, order processing, and notification services. Each service operates independently and communicates with others through RESTful APIs, ensuring flexibility and scalability.

The system is developed using Node.js and Express.js for backend services, while MongoDB is used for data storage. Docker is utilized for containerization, allowing easy deployment and consistent execution across different environments. This architecture enables the application to handle increasing user demands efficiently while simplifying maintenance and updates.

The online book store allows users to register, log in securely, browse available books, place orders, and track their purchases. Administrators can manage book inventory, monitor orders, and ensure smooth functioning of the system. The modular nature of microservices ensures that any changes or updates can be made without affecting the entire system.

Overall, this project demonstrates the implementation of a scalable and efficient online book store using modern technologies and architectural practices. It highlights the importance of distributed systems in building reliable and high-performance applications in today's digital world.

Chapter 2

PROBLEM STATEMENT

In today's digital era, online book stores are becoming increasingly popular due to their convenience and accessibility. However, many existing online bookstore systems are built using monolithic architecture, where all functionalities such as user management, book catalog, order processing, and notifications are tightly integrated into a single system. This approach leads to several challenges.

Firstly, monolithic systems are difficult to scale. As the number of users and transactions increases, the entire application must be scaled, which is inefficient and resource-intensive. Secondly, maintaining and updating such systems becomes complex, as even small changes require redeploying the entire application, increasing the risk of system failure.

Additionally, these systems often lack flexibility, making it difficult to integrate new features or adopt modern technologies. A failure in one module can affect the entire system, leading to downtime and poor user experience. Furthermore, handling high traffic and ensuring smooth performance becomes challenging in a single unified system. There is also a need for better modularity and independent service management to improve development efficiency and system reliability. Without proper separation of functionalities, debugging and testing become more complicated.

Therefore, there is a need to design and implement a scalable, flexible, and efficient online book store system that overcomes these limitations. A microservices-based architecture can address these issues by dividing the system into independent services, enabling better performance, easier maintenance, and improved scalability.

Chapter 3

OBJECTIVES OF THE PROJECT

1. **To develop an online book store using microservices architecture**
   Design and implement a scalable system where different functionalities are divided into independent services.
2. **To implement separate services for core functionalities**
   Develop individual microservices such as User Service, Book Service, Order Service, and Notification Service.
3. **To enable efficient communication using RESTful APIs**
   Ensure smooth interaction between services through well-defined API endpoints.
4. **To provide secure user authentication and authorization**
   Implement secure login and registration using encryption techniques like Bcrypt.
5. **To design a user-friendly interface**
   Create an intuitive platform for users to browse books, place orders, and track their purchases easily.
6. **To manage book inventory effectively**
   Allow administrators to add, update, and remove books from the catalog.
7. **To implement order management system**
   Enable users to place orders and track their order status efficiently.
8. **To ensure scalability and flexibility**
   Use microservices architecture to allow independent scaling and easy updates.
9. **To deploy the system using containerization**
   Use Docker to ensure consistent deployment and environment management.

Chapter-4

SYSTEM ARCHITECTURE

The **IBM Online Book Store Using Microservices** is designed using a **microservices architecture**, where the application is divided into multiple independent services. Each service is responsible for a specific functionality and communicates with other services through RESTful APIs. This architecture improves scalability, flexibility, and maintainability.

4.1 Architectural Overview

The system consists of a frontend interface, an API Gateway, multiple backend microservices, and a database. The API Gateway acts as a central entry point that routes client requests to the appropriate microservices.

4.2 Components of the System

1. Frontend (Client Layer)

The frontend provides the user interface for customers and administrators. It allows users to browse books, register, log in, place orders, and track their purchases.

2. API Gateway

The API Gateway handles all incoming requests from the frontend and routes them to the appropriate microservices. It also manages request handling, security, and response aggregation.

3. Microservices Layer

- **User Service**
  Handles user registration, login, and authentication.
- **Book Service**
  Manages book details such as title, author, price, and availability.
- **Order Service**
  Processes customer orders and tracks order status.
- **Notification Service**
  Sends notifications related to order updates and system events.

Each microservice operates independently and can be developed, deployed, and scaled separately.

4. Database Layer

MongoDB is used as the database to store user data, book information, and order details. It ensures efficient data management and retrieval.

4.3 Working of the System

1. The user interacts with the frontend interface.
2. The request is sent to the API Gateway.
3. The API Gateway forwards the request to the relevant microservice.
4. The microservice processes the request and interacts with the database.
5. The response is sent back through the API Gateway to the user.

4.4 Key Features of Architecture

- **Loose Coupling:** Services are independent of each other
- **Scalability:** Each service can be scaled individually
- **Fault Isolation:** Failure in one service does not affect others
- **Flexibility:** Easy to update or add new features

Chapter-5

TECHNOLOGIES USED

The development of the **IBM Online Book Store Using Microservices** involves the use of modern technologies for building a scalable, efficient, and maintainable system. The following technologies are used:

5.1 Node.js

Node.js is used as the backend runtime environment. It allows building fast and scalable server-side applications using JavaScript and supports asynchronous operations.

5.2 Express.js

Express.js is a lightweight web framework for Node.js used to build RESTful APIs and handle routing, middleware, and server logic efficiently.

5.3 MongoDB

MongoDB is a NoSQL database used to store application data such as user details, book information, and order records. It provides flexibility and high performance.

5.4 Mongoose

Mongoose is an Object Data Modeling (ODM) library used to interact with MongoDB. It helps in defining schemas and managing data effectively.

5.5 Docker

Docker is used for containerization of microservices. It ensures that each service runs in an isolated environment, making deployment consistent and reliable.

5.6 Docker Compose

Docker Compose is used to manage and run multiple containers together. It simplifies the process of running all microservices as a single application.

5.7 Bcrypt

Bcrypt is used for secure password hashing. It ensures that user passwords are stored in an encrypted format, improving security.

5.8 RESTful APIs

REST APIs are used for communication between microservices. They enable smooth data exchange and integration between different components.

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Chapter-6

METHODOLOGY

The development of the **IBM Online Book Store Using Microservices** follows a systematic and structured approach to ensure efficient design, development, and deployment of the system. The methodology consists of the following phases:

6.1 Requirement Analysis

In this phase, the requirements of the system are identified and analyzed. The main functionalities include user registration and login, browsing books, managing inventory, placing orders, and tracking order status. Both user and administrator requirements are considered.

6.2 System Design

The system is designed using a microservices architecture. The application is divided into independent services such as User Service, Book Service, Order Service, and Notification Service. The interactions between services are defined using RESTful APIs, and the overall system flow is planned.

6.3 Development Phase

Each microservice is developed independently using Node.js and Express.js. Separate modules are created for handling different functionalities, including routing, controllers, and database operations. MongoDB is used for storing data.

6.4 Integration of Services

After developing individual services, they are integrated using an API Gateway. The API Gateway routes client requests to the appropriate microservice and ensures smooth communication between components.

6.5 Testing

Testing is performed to ensure that the system works correctly and efficiently:

- **Unit Testing:** Testing individual components
- **API Testing:** Verifying endpoints using tools like Postman
- **Integration Testing:** Ensuring proper communication between services

6.6 Deployment

The system is deployed using Docker and Docker Compose. Each microservice runs in a separate container, ensuring consistent and reliable deployment across different environments.

6.7 Maintenance and Improvement

After deployment, the system is monitored for performance and errors. Necessary updates and improvements are made to enhance functionality and user experience.

CHAPTER 7

IMPLEMENTATION

The implementation of the **IBM Online Book Store Using Microservices** involves developing and integrating multiple independent services to build a complete and functional system. Each service is designed to perform a specific task and is deployed independently, ensuring flexibility and scalability.

7.1 Development Environment Setup

The project is developed using Node.js as the runtime environment and Express.js for building server-side applications. MongoDB is configured as the database, and necessary dependencies are installed using npm. Docker is used to containerize the services for smooth deployment.

7.2 Frontend Implementation

The frontend provides an interactive user interface for both customers and administrators.

Key Features:

- User registration and login
- Book browsing interface
- Add to cart and order placement
- Order tracking dashboard
- Admin panel for managing books

The frontend communicates with backend services through REST APIs and ensures a smooth user experience.

7.3 Backend Microservices Implementation

The backend is divided into multiple microservices, each responsible for a specific functionality.

1. User Service

- Handles user registration and login
- Encrypts passwords using Bcrypt
- Manages authentication

2. Book Service

- Performs CRUD operations on books
- Stores details such as title, author, price, and stock

3. Order Service

- Manages order creation and updates
- Tracks order status (Placed, Shipped, Delivered)

4. Notification Service

- Sends notifications to users about order updates

Each microservice runs on a separate port and has its own routes and controllers.

7.4 API Gateway Implementation

An API Gateway is implemented to act as a central entry point. It receives all client requests and forwards them to the appropriate microservices. This improves security and simplifies communication between the frontend and backend.

7.5 Database Implementation

MongoDB is used to store application data:

- User details
- Book catalog
- Order information

Mongoose is used to define schemas and manage database operations efficiently.

7.6 Authentication and Security

User authentication is implemented using secure techniques:

- Password hashing using Bcrypt
- Token-based authentication (JWT)
- Protected API routes

This ensures that only authorized users can access specific functionalities.

7.7 Functional Workflow

1. User registers and logs in
2. User browses available books
3. User places an order
4. Order Service processes the request
5. Notification Service sends updates
6. User tracks order status

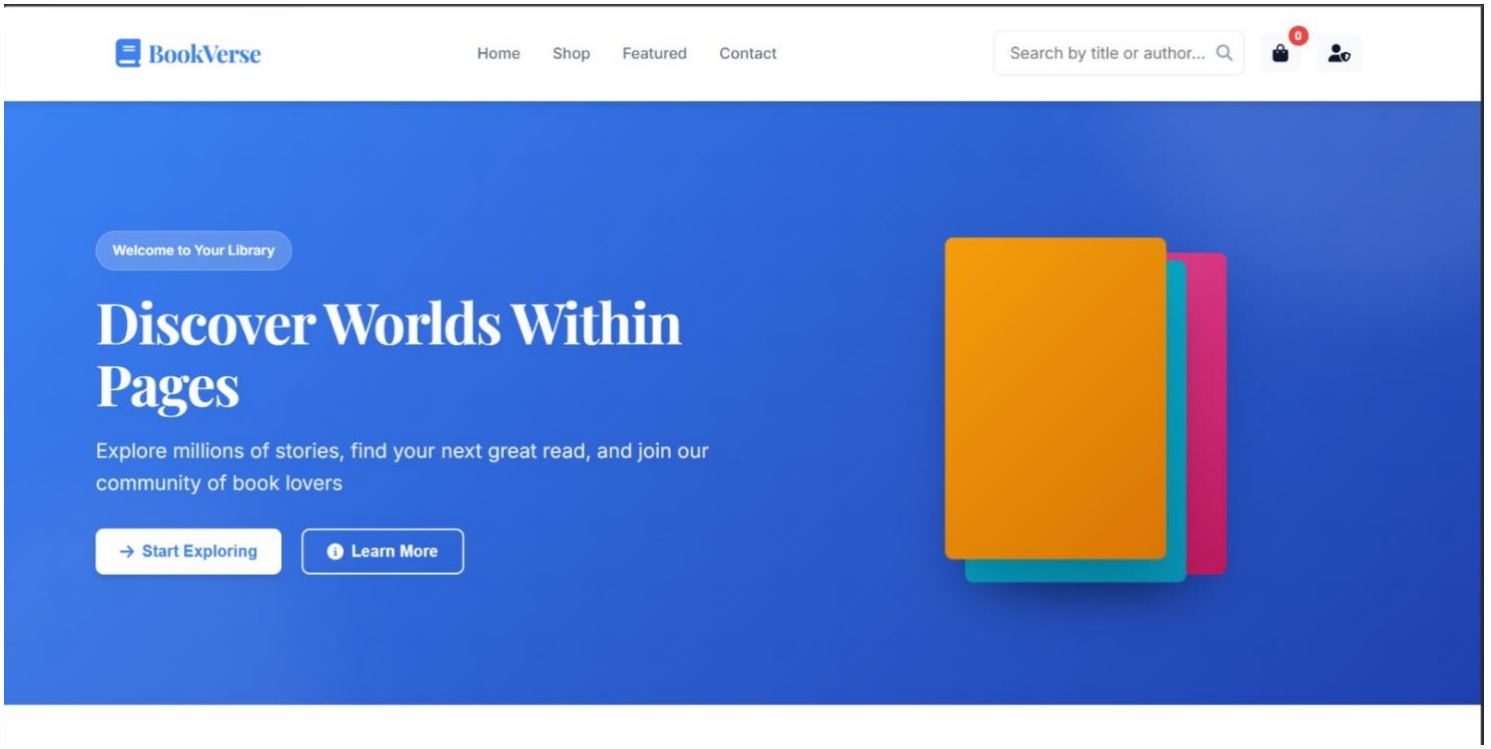
7.8 Testing and Debugging

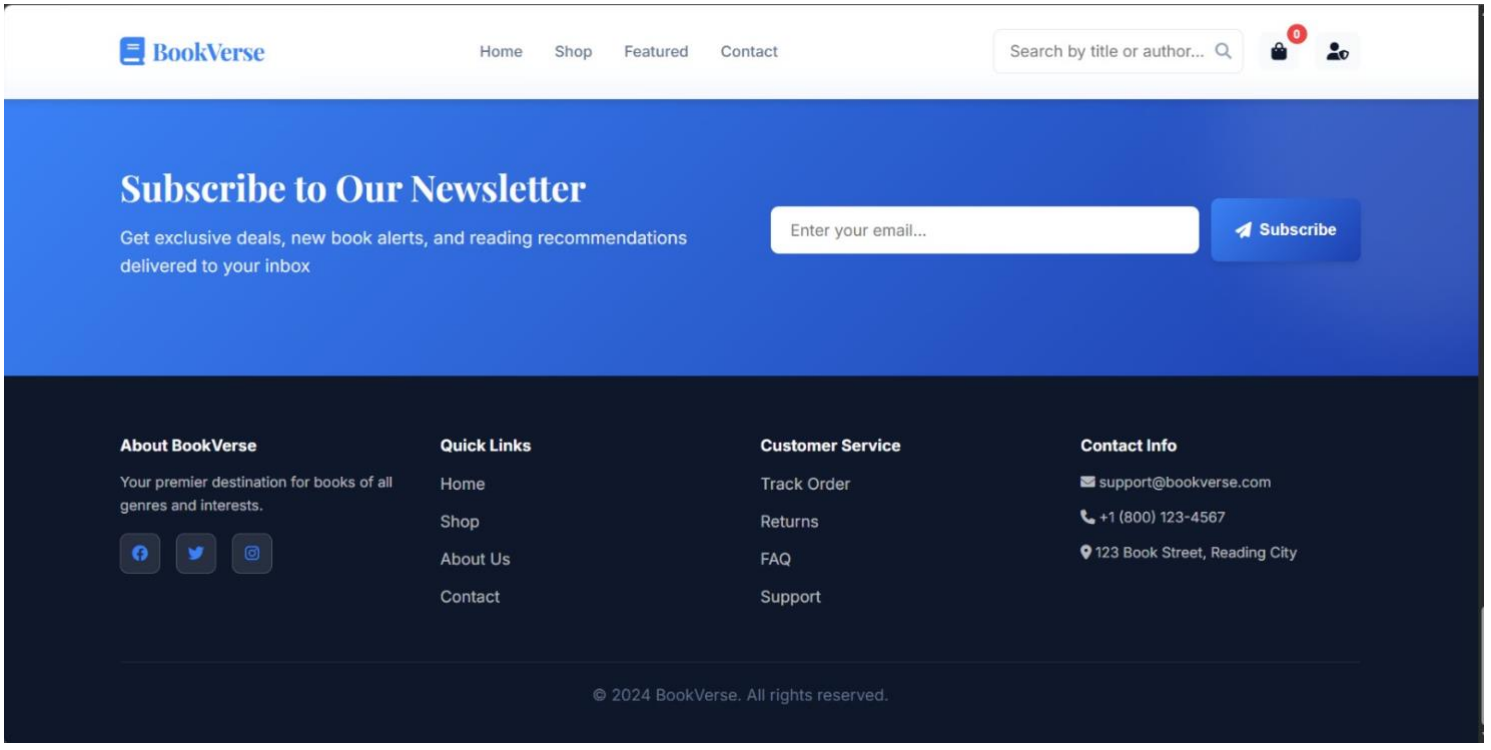
The system is tested to ensure proper functionality:

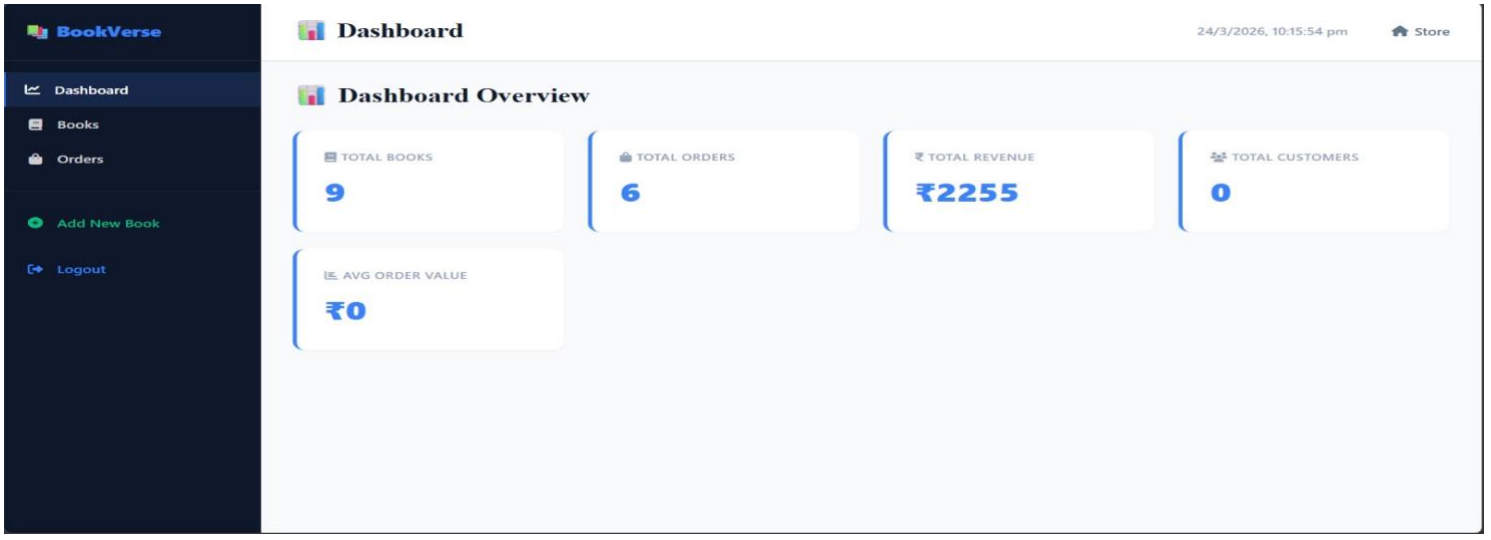
- API testing using Postman
- Debugging errors in services
- Verifying data flow between services

Chapter 8

RESULT



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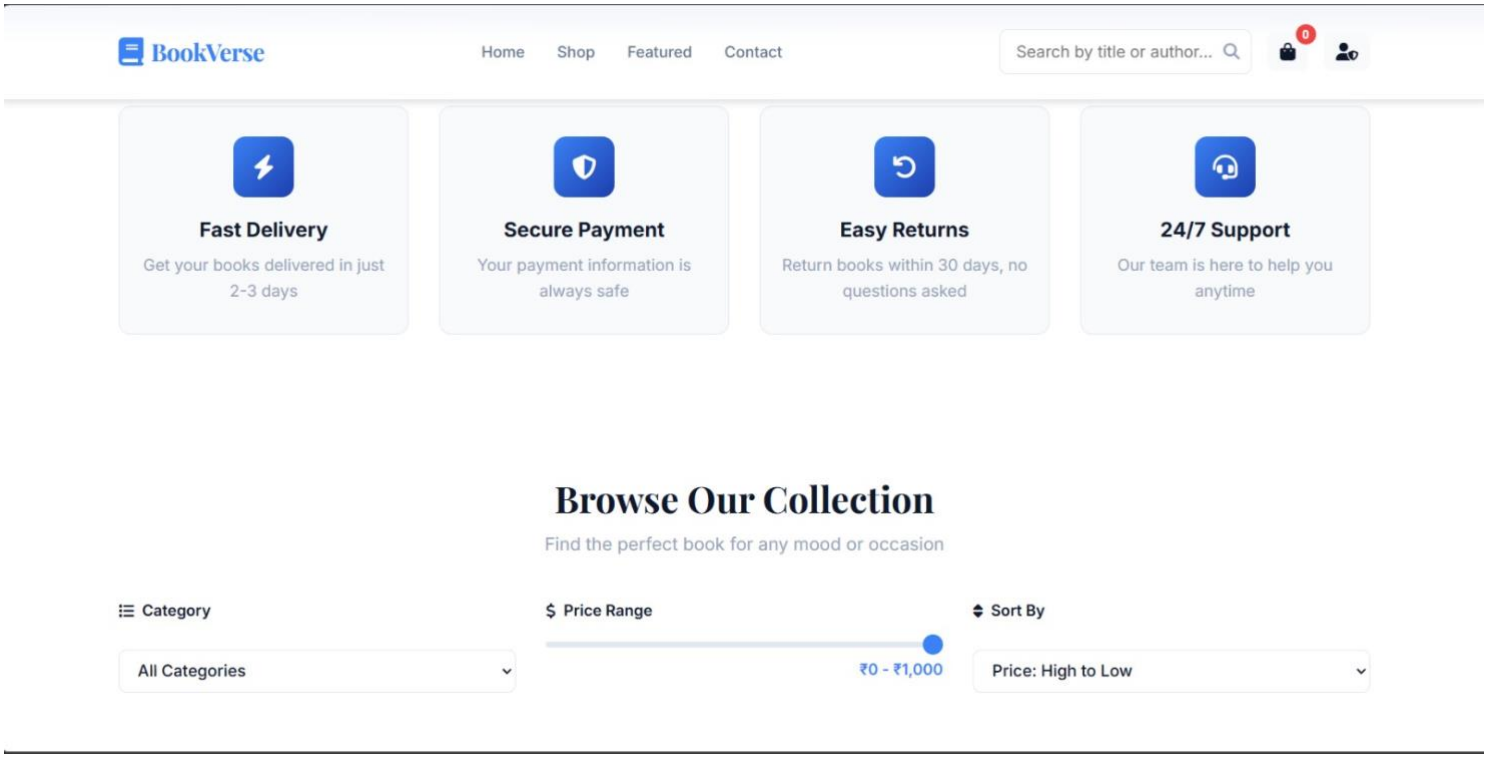
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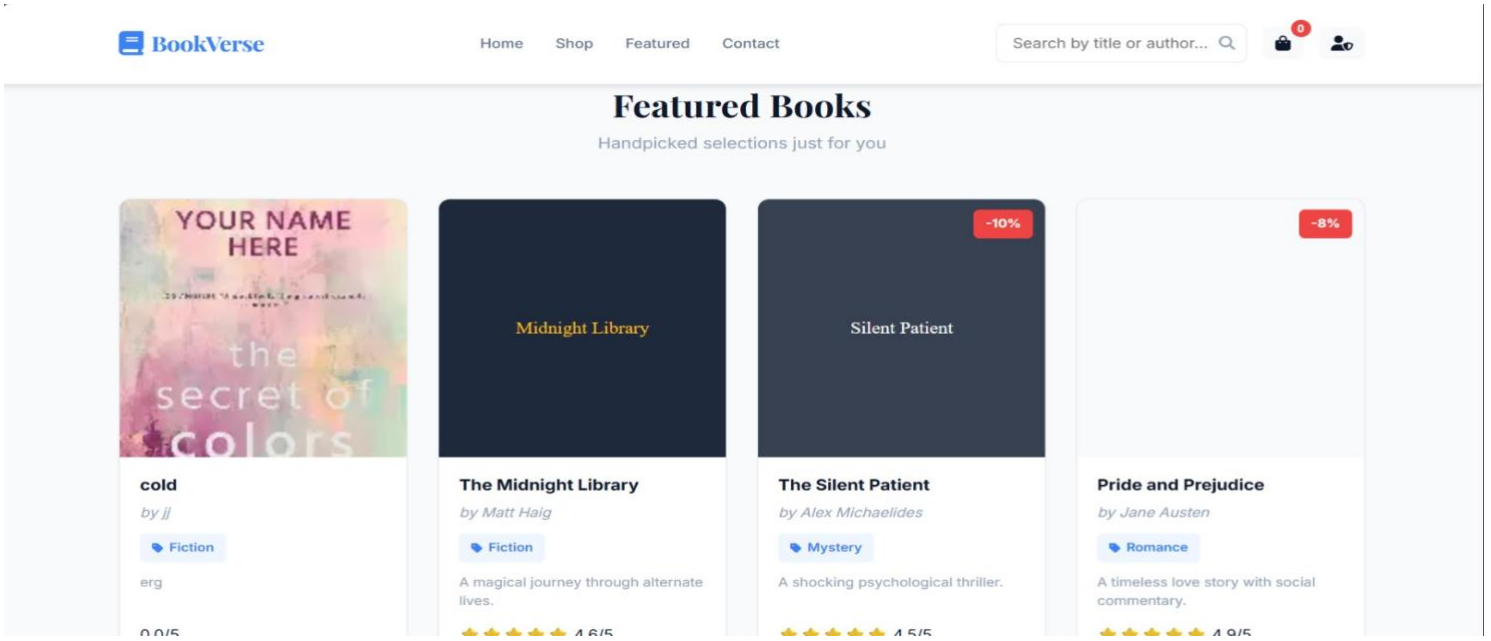
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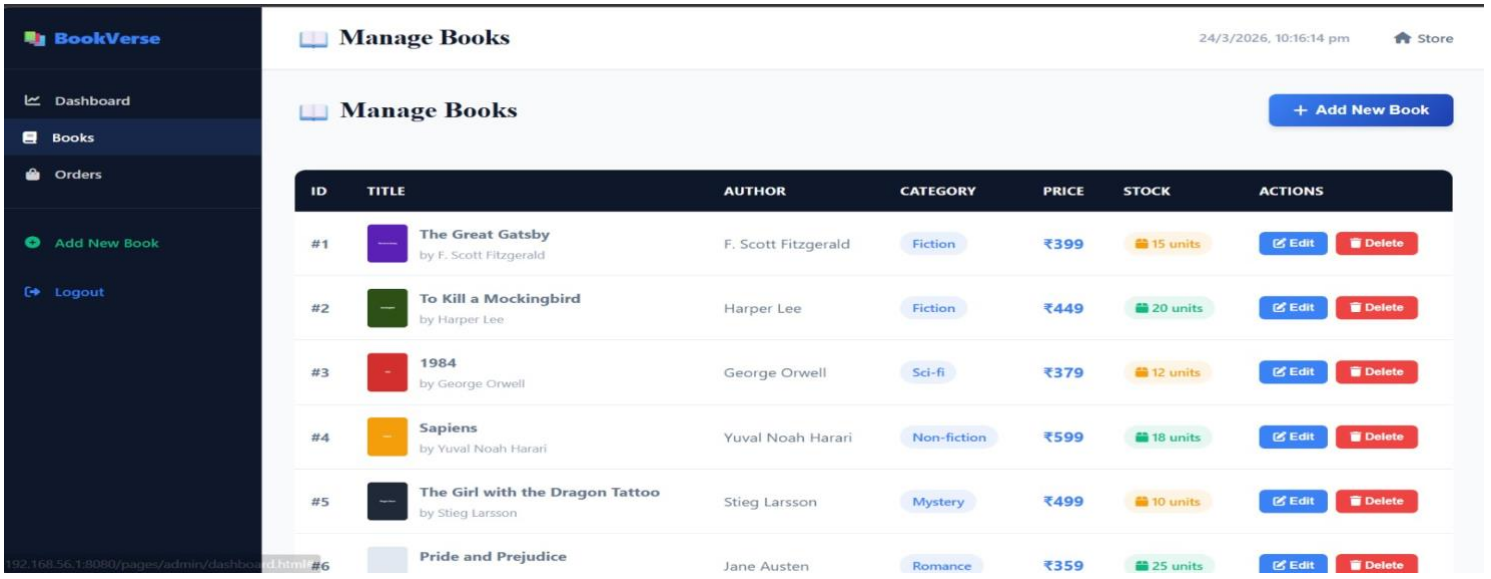
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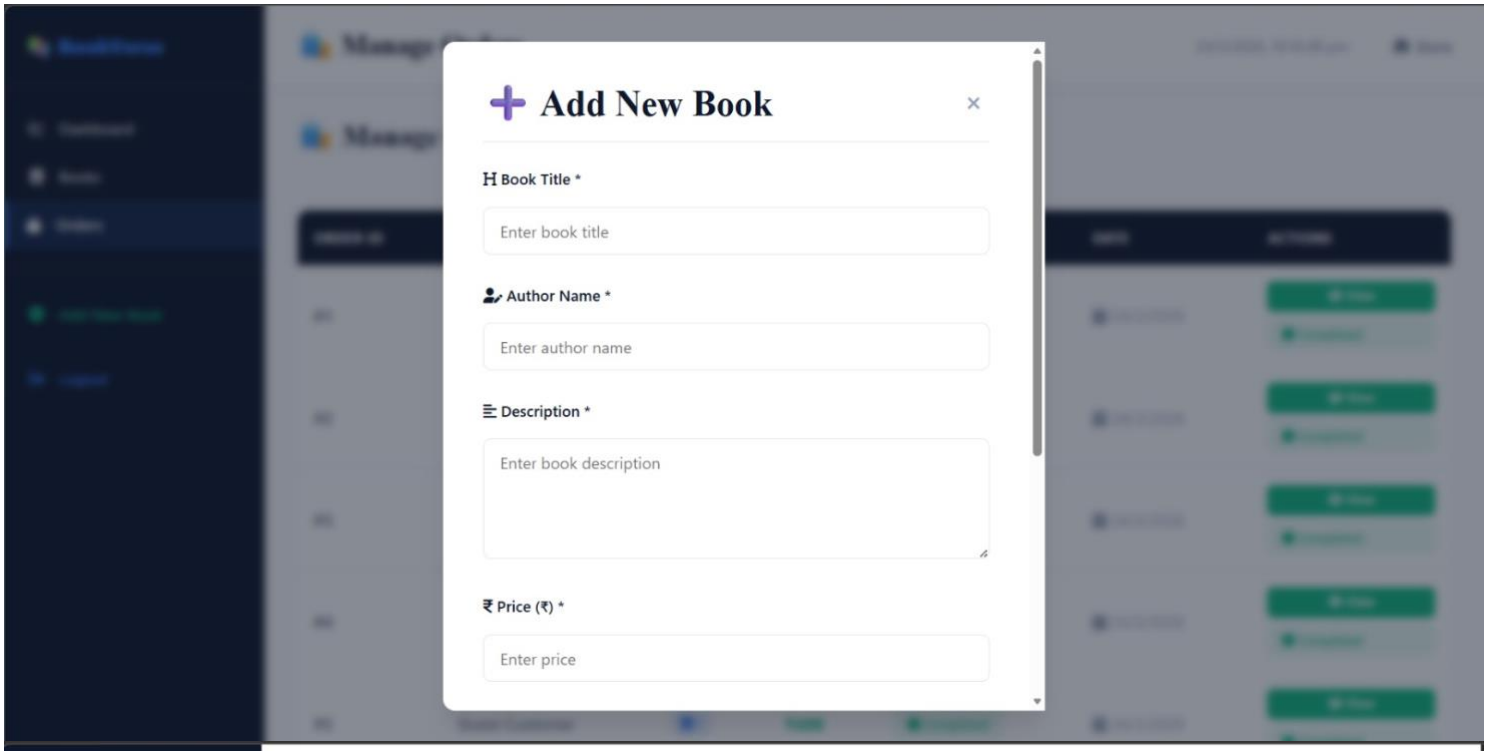
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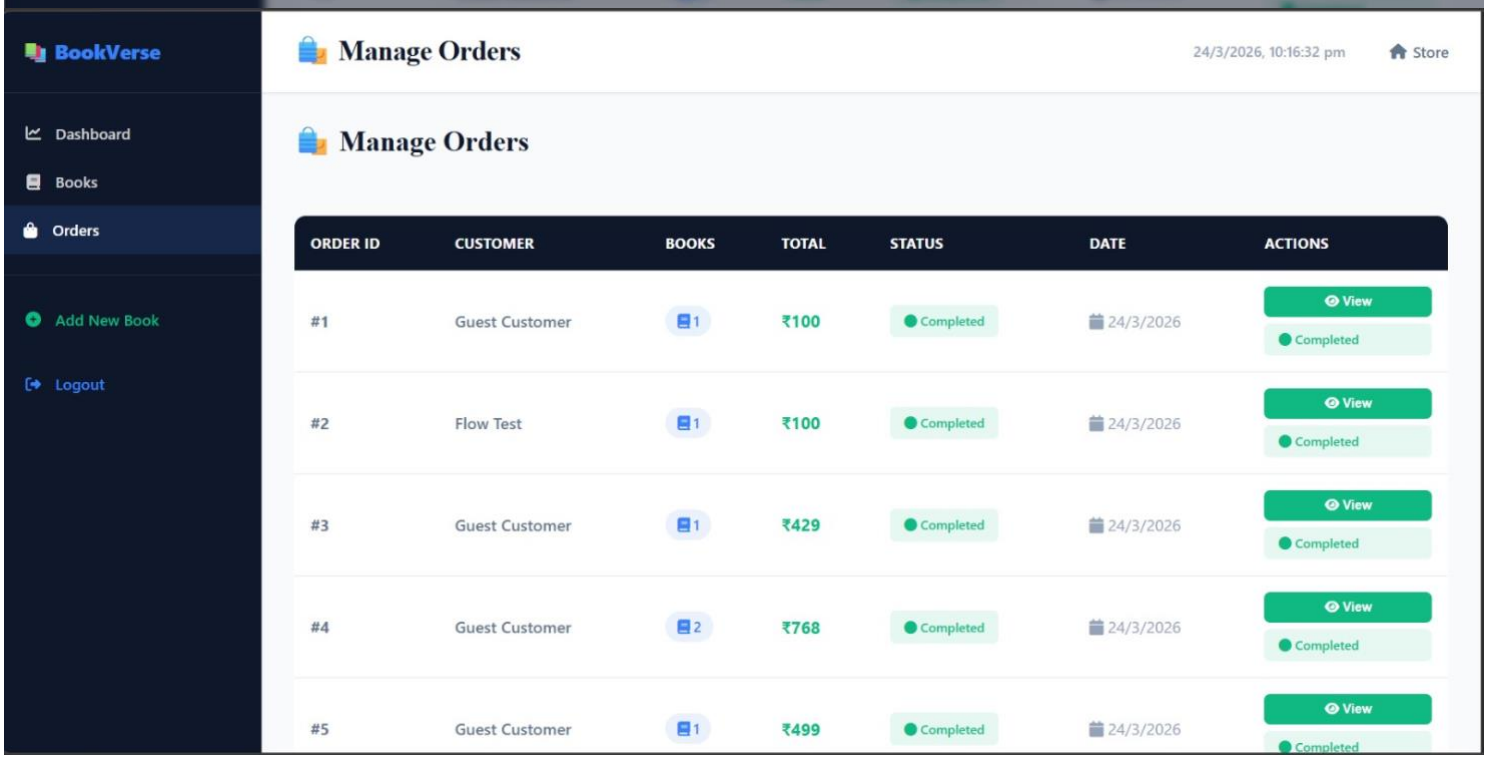
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Chapter-9

ADVANTAGES & LIMITATIONS

9.1 Advantages

1. Scalability

   The microservices architecture allows each service to scale independently based on demand, improving overall system performance.
2. Flexibility and Modularity

   Each module (User, Book, Order, Notification) is developed independently, making the system easy to modify and extend.
3. Improved Fault Isolation

   Failure in one service does not affect the entire system, ensuring better reliability.
4. Faster Development and Deployment

   Multiple services can be developed and deployed simultaneously, reducing development time.
5. Technology Independence

   Different services can use different technologies if required, providing flexibility in development.
6. Better Maintainability

   Smaller and independent services are easier to maintain, debug, and update.
7. Efficient Resource Utilization

   Only required services can be scaled, reducing unnecessary resource usage.

9.2 Limitations

1. Increased Complexity

   Managing multiple services is more complex compared to a monolithic architecture.
2. Difficult Debugging

   Identifying issues across multiple services can be challenging.
3. Network Dependency

   Communication between services depends on network calls, which may introduce latency.
4. Deployment Challenges

   Requires proper configuration and management of multiple containers and services.
5. Data Management Complexity

   Handling data consistency across different services can be difficult.
6. Higher Initial Setup Effort

   Setting up microservices architecture requires more planning and configuration.

Chapter-10

FUTURE ENHANCEMENT

The **IBM Online Book Store Using Microservices** can be further improved by adding advanced features and enhancing system capabilities. The following enhancements are proposed:

10.1 Payment Gateway Integration

Integrate secure online payment options such as credit/debit cards, UPI, and net banking to enable real-time transactions.

10.2 Recommendation System

Implement an AI-based recommendation system to suggest books based on user preferences, purchase history, and browsing behavior.

10.3 Search and Filtering Improvements

Enhance the search functionality by adding advanced filters such as category, author, price range, and ratings.

10.4 User Reviews and Ratings

Allow users to provide feedback, ratings, and reviews for books, improving user engagement and decision-making.

10.5 Mobile Application Development

Develop a mobile app (Android/iOS) to provide better accessibility and user experience.

10.6 Real-Time Notifications

Improve the notification system using technologies like WebSockets for real-time order updates.

10.7 Cloud Deployment

Deploy the system on cloud platforms such as AWS or Azure for better scalability, availability, and performance.

10.8 Advanced Security Features

Implement features such as two-factor authentication (2FA), role-based access control, and data encryption for enhanced security.

10.9 Analytics Dashboard

Provide an admin dashboard with analytics such as sales reports, user activity, and popular books.

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Chapter-11

CONCLUSION

The **IBM Online Book Store Using Microservices** project successfully demonstrates the design and development of a scalable, flexible, and efficient e-commerce platform. By adopting a microservices architecture, the system divides core functionalities such as user management, book catalog, and order processing into independent services, improving maintainability and performance.

The application provides a user-friendly interface that allows users to browse books, place orders, and track their purchases بسهولة. The use of modern technologies like Node.js, Express.js, MongoDB, and Docker ensures smooth communication between services and reliable system performance.

This project achieves its primary objectives by delivering a modular and secure system capable of handling real-world requirements. It also highlights the advantages of microservices in terms of scalability, fault isolation, and ease of deployment.

In conclusion, the system serves as a strong foundation for building advanced online bookstore platforms and can be further enhanced with additional features to meet evolving user needs and industry standards.

Chapter-12

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- **GitHub Guides. Available: https://github.com/pdprajwal9522-cloud/ibm_project.git**
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